Cut Sheet

Board Design Summary

Style: checkerboard
Dimensions: 12" (305 mm) × 18" (457 mm) × 1-1/2" (38 mm)
Species: Walnut, Maple

Cut List

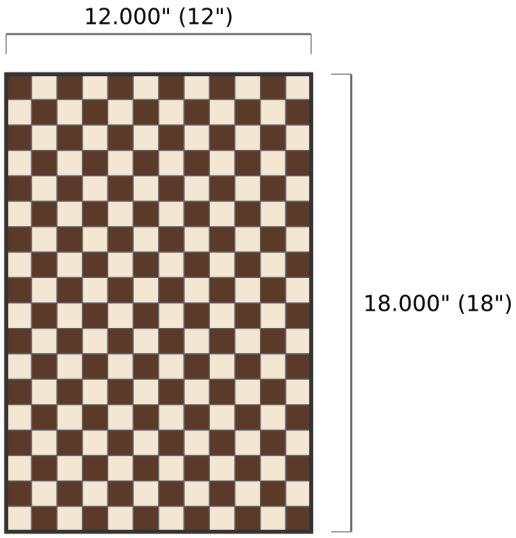
Part	Width	Thickness	Length	Species
Board strips	1.000" (1")	1.500" (1 1/2")	18.000" (18")	Walnut

Cut List

Part	Qty	Width	Thickness	Length	Species
Panel A, sub-panel 1/2: Walnut/Maple strip-set panel: glue 6 whole strips (order: W M W M W M), rip each strip 1-1/16" (27 mm), then joint to 1" (25 mm) wide; strips milled 1-1/16" (27 mm) thick; plane the glued panel to the square size 1" (25 mm); Strip-set (long grain, before slicing): Planer OK: 6-1/4" (158 mm) wide, long grain; then crosscut sled with stop block (never fence) 18 slices × 1-11/16" (43 mm) each (sled must clear 6-1/4" (158 mm)); Too wide for the jointer (6" (152 mm)): flatten one face on a planer sled first, then plane the other face parallel	1	6-1/4" (158 mm)	1" (25 mm)	34-3/4" (883 mm)	Walnut/Maple

Panel A, sub-panel 2/2: Walnut/Maple strip-set panel: glue 6 whole strips (order: W M W M W M), rip each strip 1-1/16" (27 mm), then joint to 1" (25 mm) wide; strips milled 1-1/16" (27 mm) thick; plane the glued panel to the square size 1" (25 mm); Strip-set (long grain, before slicing): Planer OK: 6-1/4" (158 mm) wide, long grain; then crosscut sled with stop block (never fence) 18 slices × 1-11/16" (43 mm) each (sled must clear 6-1/4" (158 mm)); Too wide for the jointer (6" (152 mm)): flatten one face on a planer sled first, then plane the other face parallel	1	6-1/4" (158 mm)	1" (25 mm)	34-3/4" (883 mm)	Walnut/Maple

Board Layout (Top View)



Step 1: Mill rough stock

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3.0 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Settings:

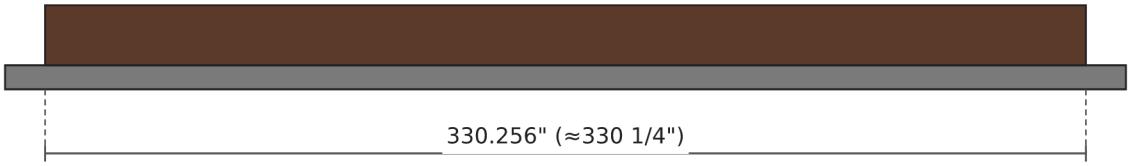
Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.7 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

Checkerboard (End Grain): step 1: Mill rough stock

Whole blank on the jointer (schematic, not to scale)



Mill and rip whole blanks before any crosscut. Blank length = max(305 mm, slices x (slice + kerf) + 2 x 25 mm trim). Never send a crosscut slice or piece shorter than 305 mm to the jointer or planer.

Walnut strip: blank 330.256" (≈330 1/4") long, 102 slices of 3.094" (≈3 3/32") crosscut after milling.

Maple strip: blank 330.256" (≈330 1/4") long, 102 slices of 3.094" (≈3 3/32") crosscut after milling.

Blank lengths: catalog blankRule applied to the app cut list by boardviz; the app export does not apply the rule yet.

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.5 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

Step 2: Rip strips

Rip strips with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

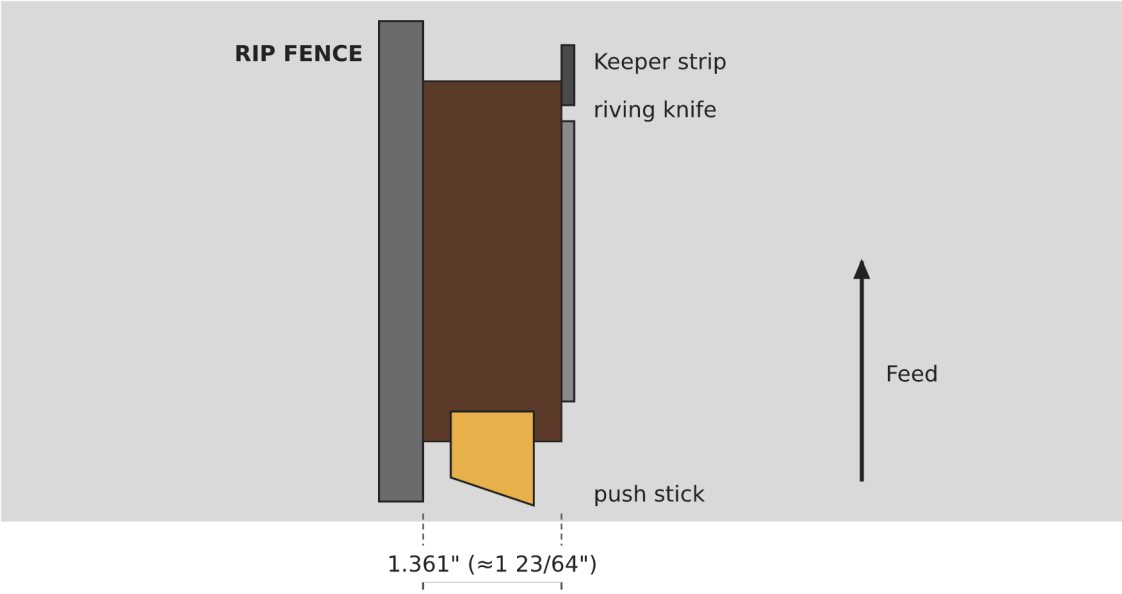
Settings:

Fence to blade: 1-1/16" (27 mm)

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

Checkerboard (End Grain): rip-fence setup, step 2: Rip strips



Measure fence face → near side of a tooth

set toward the fence

Kerf: 1/8" (3.2 mm) full-kerf blade

Rip strips with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

Fence to blade: strip width from cut list

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

Step 3: Glue up

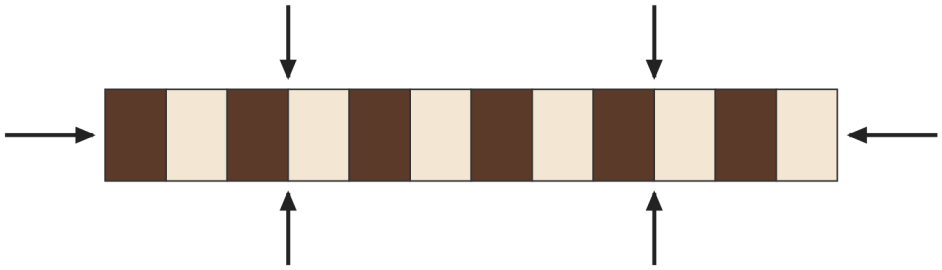
Glue the first edge-grain panel. Alternate clamps above and below the panel and check for flat with a straightedge.

Settings:

Clamp spacing: about 8" (203 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

Checkerboard (End Grain): glue-up 1, step 3: Glue up



Glue the first edge-grain panel. Alternate clamps above and below the panel and check for flat with a straightedge.

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

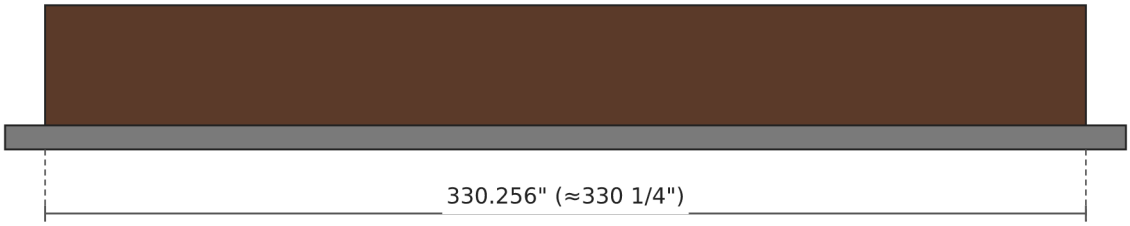
Step 4: Flatten first panel

Flatten Panel A, sub-panel 1/2, 6-1/4" (158 mm) wide:
Strip-set (long grain, before slicing): Planer OK: 6-1/4" (158 mm) wide, long grain.
Flatten Panel A, sub-panel 2/2, 6-1/4" (158 mm) wide:
Strip-set (long grain, before slicing): Planer OK: 6-1/4" (158 mm) wide, long grain.

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

Checkerboard (End Grain): step 4: Flatten first panel

Panel on the planer (schematic, not to scale)



Mill and rip whole blanks before any crosscut. Blank length = max(305 mm, slices x (slice + kerf) + 2 x 25 mm trim). Never send a crosscut slice or piece shorter than 305 mm to the jointer or planer.

Walnut strip: blank 330.256" (≈330 1/4") long, 102 slices of 3.094" (≈3 3/32") crosscut after milling.

Maple strip: blank 330.256" (≈330 1/4") long, 102 slices of 3.094" (≈3 3/32") crosscut after milling.

Blank lengths: catalog blankRule applied to the app cut list by boardviz; the app export does not apply the rule yet.

Plane or sand the panel flat. It is still long grain, so the planer is allowed here if the panel is no wider than 12" (305 mm); a wider panel goes to the router sled or drum sander.

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

Step 5: Crosscut slices

Crosscut the panel into slices of the final board thickness plus the end-grain flattening allowance (5 mm), on a crosscut sled with a stop block.
Slice size: 1-11/16" (43 mm).
Final thickness: 1-1/2" (38 mm).
End-grain allowance: 3/16" (5.0 mm).
Slices: 36 in all (Panel A, sub-panel 1/2 18, Panel A, sub-panel 2/2 18).
Panel A, sub-panel 1/2 (order: W M W M W M): sled must clear 6-1/4" (158 mm).
Panel A, sub-panel 2/2 (order: W M W M W M): sled must clear 6-1/4" (158 mm).

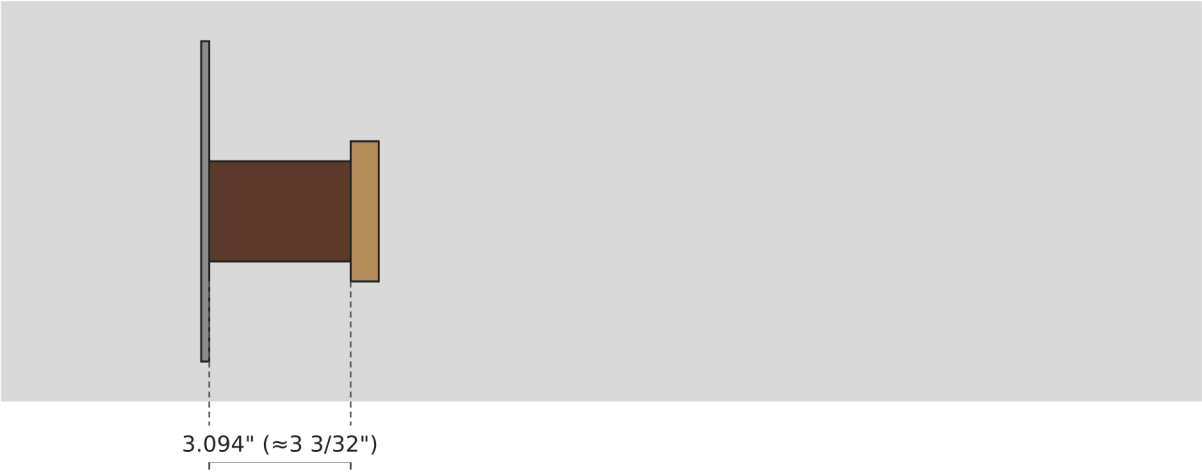
Settings:

Slice length: 1-11/16" (43 mm)

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

Checkerboard (End Grain): crosscut, step 5: Crosscut slices



Crosscut the panel into slices of the final thickness on a crosscut sled with a stop block.

Slice length: final board thickness from cut list

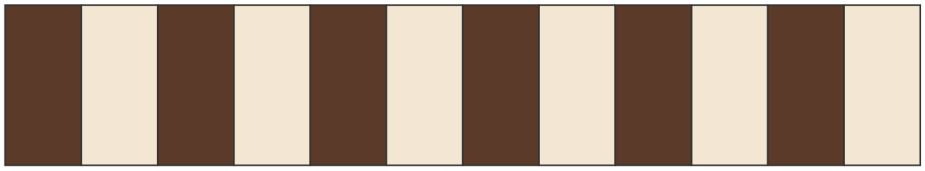
DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

Step 6: Rotate slices

Turn each slice end grain up.

Checkerboard (End Grain): strip sequence



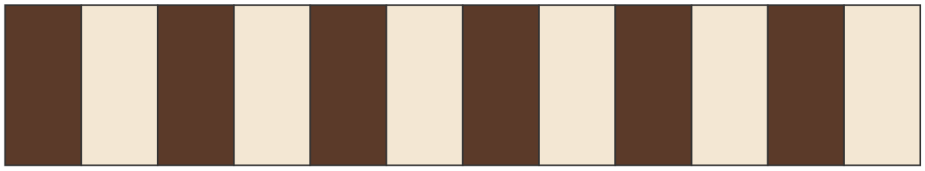
1. Walnut
2. Maple
3. Walnut
4. Maple
5. Walnut
6. Maple
7. Walnut
8. Maple
9. Walnut
10. Maple
11. Walnut
12. Maple

Turn each slice end grain up.

Step 7: Flip alternate slices

Flip every other slice end for end so the colours alternate into squares.

Checkerboard (End Grain): strip sequence



1. Walnut
2. Maple
3. Walnut
4. Maple
5. Walnut
6. Maple
7. Walnut
8. Maple
9. Walnut
10. Maple
11. Walnut
12. Maple

Flip every other slice end for end so the colours alternate into squares.

Step 8: Glue up

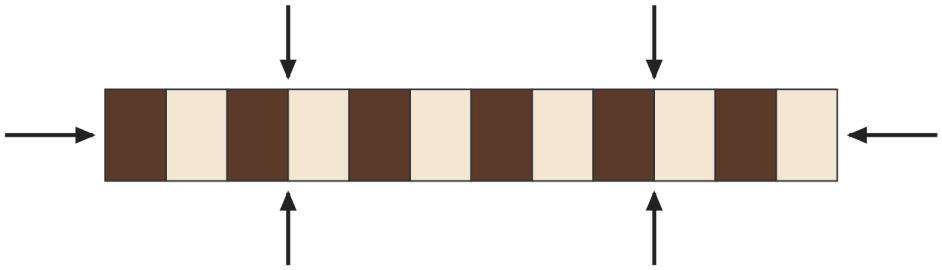
Glue slices into the final panel, aligning the squares at every joint. Alternate clamps above and below the panel and check for flat with a straightedge.

Settings:

Clamp spacing: about 8" (203 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

Checkerboard (End Grain): glue-up 2, step 8: Glue up



Glue slices into the final panel, aligning the squares at every joint. Alternate clamps above and below the panel and check for flat with a straightedge.

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

Step 9: Flatten and sand

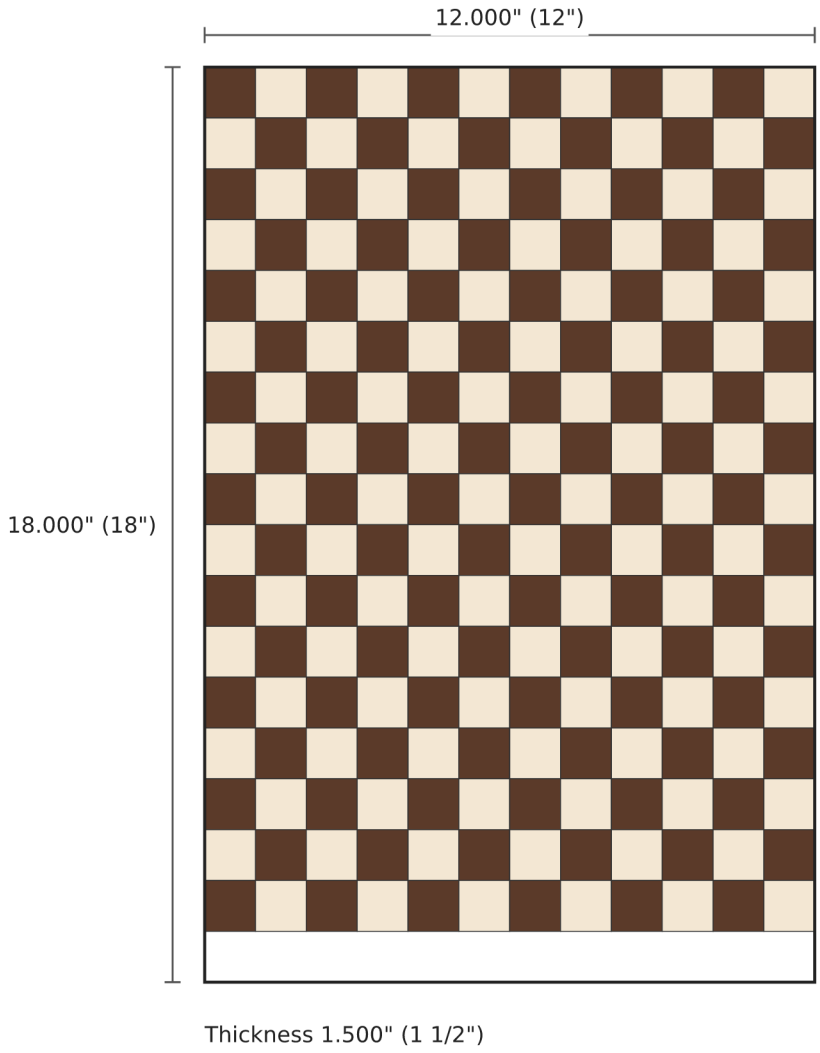
Flatten the final glue-up, 12" (305 mm) wide:
end grain: router sled or drum sander, never the planer.
Then sand: 80, 120, 180 grit. Break edges (180 grit).

Settings:

Grits: 80 / 120 / 180

DANGER: Never run an end-grain glue-up through the thickness planer. It can blow apart. Flatten with a router sled or drum sander.

Checkerboard (End Grain): top view



Flatten with a router sled or drum sander, never the thickness planer. Sand 80, 120, 180 grit. Break edges (180 grit).

Grits: 80 / 120 / 180

DANGER: Never run an end-grain glue-up through the thickness planer. It can blow apart. Flatten with a router sled or drum sander.

Step 10: Edge profile

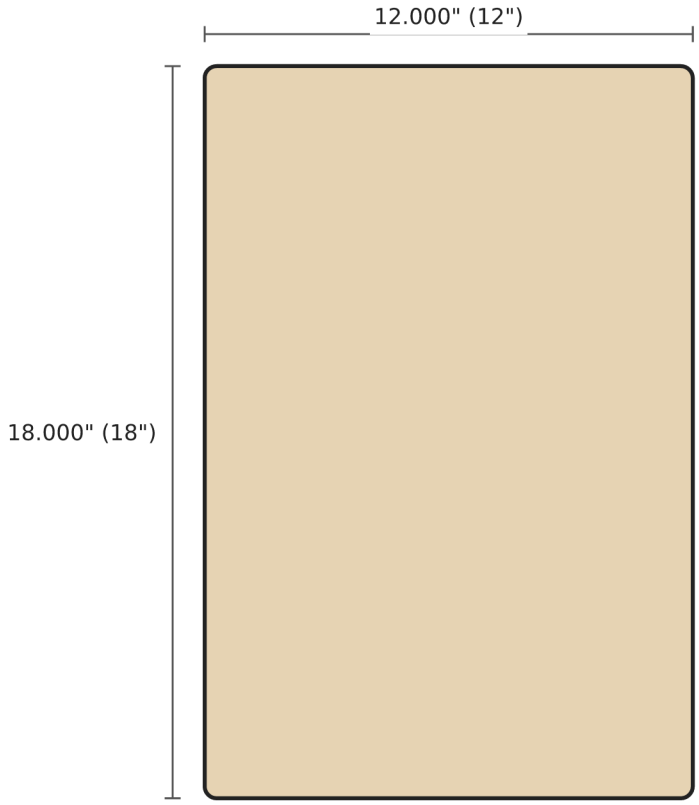
Rout the edge profile, default: 1/8" (3.2 mm) roundover. Optional finger grips: core box / round-nose bit 3/8-1/2" (9.5-12.7 mm).

Settings:

Bit: 1/8" (3.2 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

Checkerboard (End Grain): outline, step 10: Edge profile



Rout the edge profile, default: 1/8" (3 mm) roundover. Optional finger grips: core box / round-nose bit 3/8–1/2" (9.5–12.7 mm).

Bit: 1/8" (3 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

Step 11: Finish

Raise the grain with a damp cloth, sand 180 again, then flood with food-safe mineral oil and top with board butter.

NOTE: Use closed-grain, non-toxic hardwoods only. Avoid spalted, softwood or oily/sensitising exotics on cutting surfaces.